5/03

Structures

A structure is a userdefined datatype , which is collection of heterogenous elements.

syntax to declare a structure

```
struct tagname
{
        members of structure;

};
```

ex :

```
struct Product
{
        char name[20];
        char author[20];
        float price;
        int pages;
};
```

Keyword

tagname

members

creating or designing a new datatype called struct Product

Syntax to create Structure variable

```
datatype varname;
```

ex:

```
struct tagname variblename;
struct Product var;
```

datatype

variable name

```
int main()
{ int; X no memory allocated to int
=
}
```

```
int main()
{ int x;
}
```

it gets 4 byte of memory.

Points to remember:

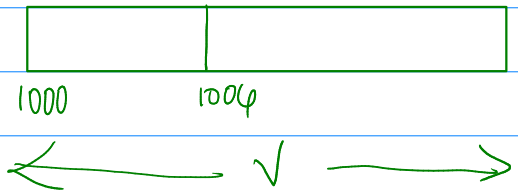
1. structure declaration does not occupy the memory, it is just a logical existence.
So no memory will be allocated to members of structure, enhance cannot be initialized inside the structure declaration.

```
struct ABC
{           int data1=10;
            char s[]="abcd";
};
```

Invalid

2. Physical existences comes with structure variable creation.
So to use the declared structure we need to create a variable of it and use it.

```
struct ABC
{
        int data1;
        char s[12];
};
```




```
struct ABC v;
```

3. structure will not represent the base address , so we cannot use index or dereference operator
then howe to access the members of structure
we need use dot . operator
ex :
to access the data1 and s of above structure we use
v.data1
v.s

4. sizeof structure is sum of the sizes of the members of structure

Variable declarations

1. variable declaration along with structure declaration.
2. variable declaration after structure declaration.

method1 :

```
struct tagname
{
        members;

}varname;
```

ex : struct ABC

```
{
        int data1;
        char s[20];
}var;
```

method2 :

```
struct tagname
{
        members;
};

struct tagname varname;
```

ex : struct ABC

```
{
        int data1;
        char s[20];
};

struct ABC var1;
```